# Predictive Analytics for the PWHL Using Monte Carlo Simulation

**Forecasting playoff probabilities in a new professional league with limited data**

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Code Repository

## Background / Problem

The Professional Women's Hockey League (PWHL), founded in 2024, has limited historical data, making traditional sports prediction models difficult to apply. Established leagues rely on decades of statistics to power analytics while emerging leagues must operate with small datasets and evolving team compositions.

This project explores whether **Monte Carlo simulation** can generate meaningful playoff probability and final standings forecasts using only current-season performance data.

## Model Design

Game-level data is collected from the **PWHL HockeyTech API** and stored in a **PostgreSQL** database using a **SQLAlchemy** pipeline. Automated workflows update the database after each game and generate visualizations for the ByTheNumbers analytics platform.

Team strength is estimated using **a lightweight feature** set optimized for low-data environments, **including points percentage (recency-weighted), a composite performance score (streak, points per game, goal differential), home win percentage, shot ratio, and an expected-goals proxy.**

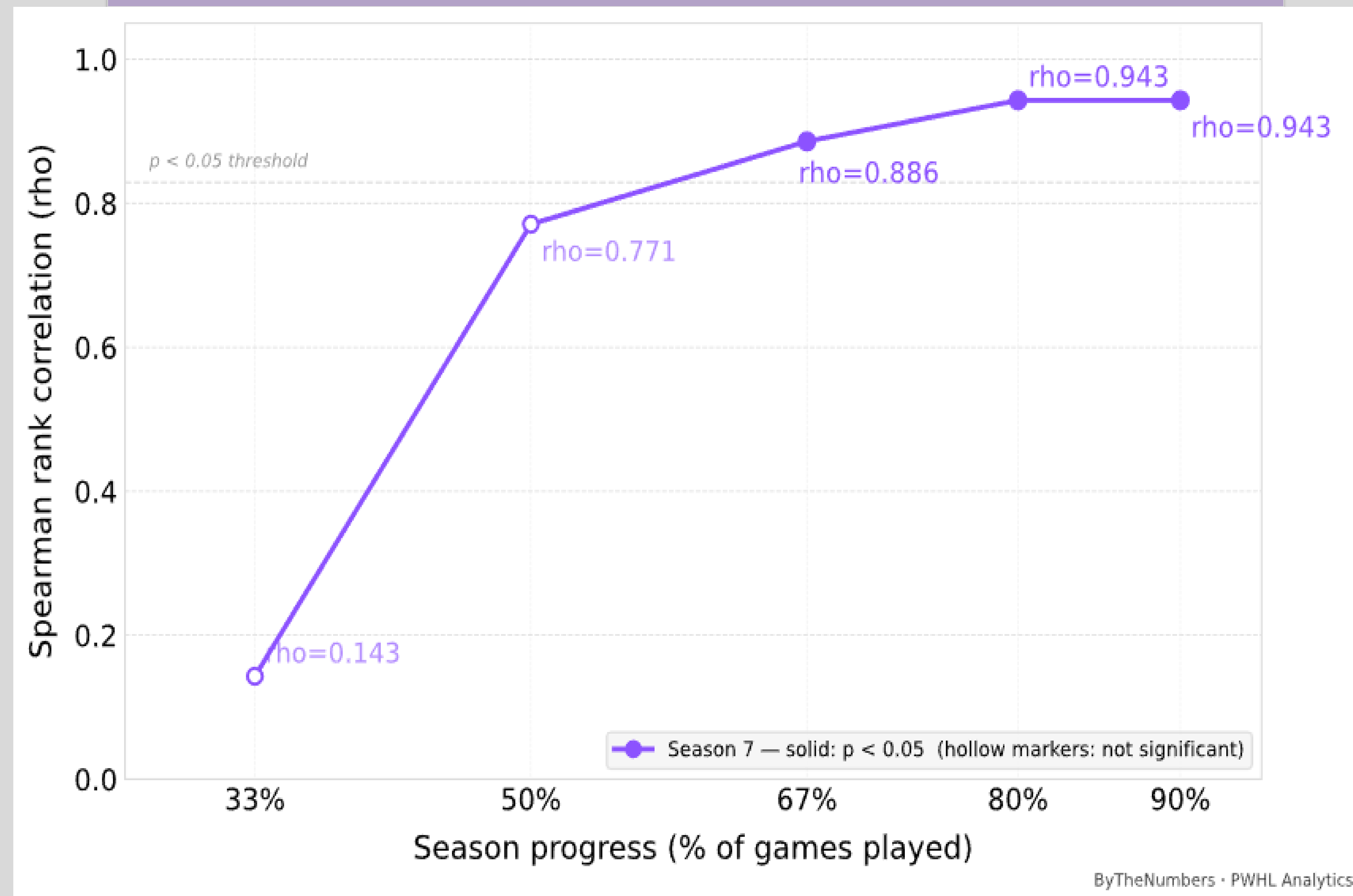
Feature importance is learned via **Ridge Regression**, with **points percentage (0.30) and composite performance score (0.28)** emerging as the strongest predictors. Notably, expected-goals and recent-form features contribute minimally, suggesting that cumulative performance metrics carry more signal than short-window form in low-data settings.

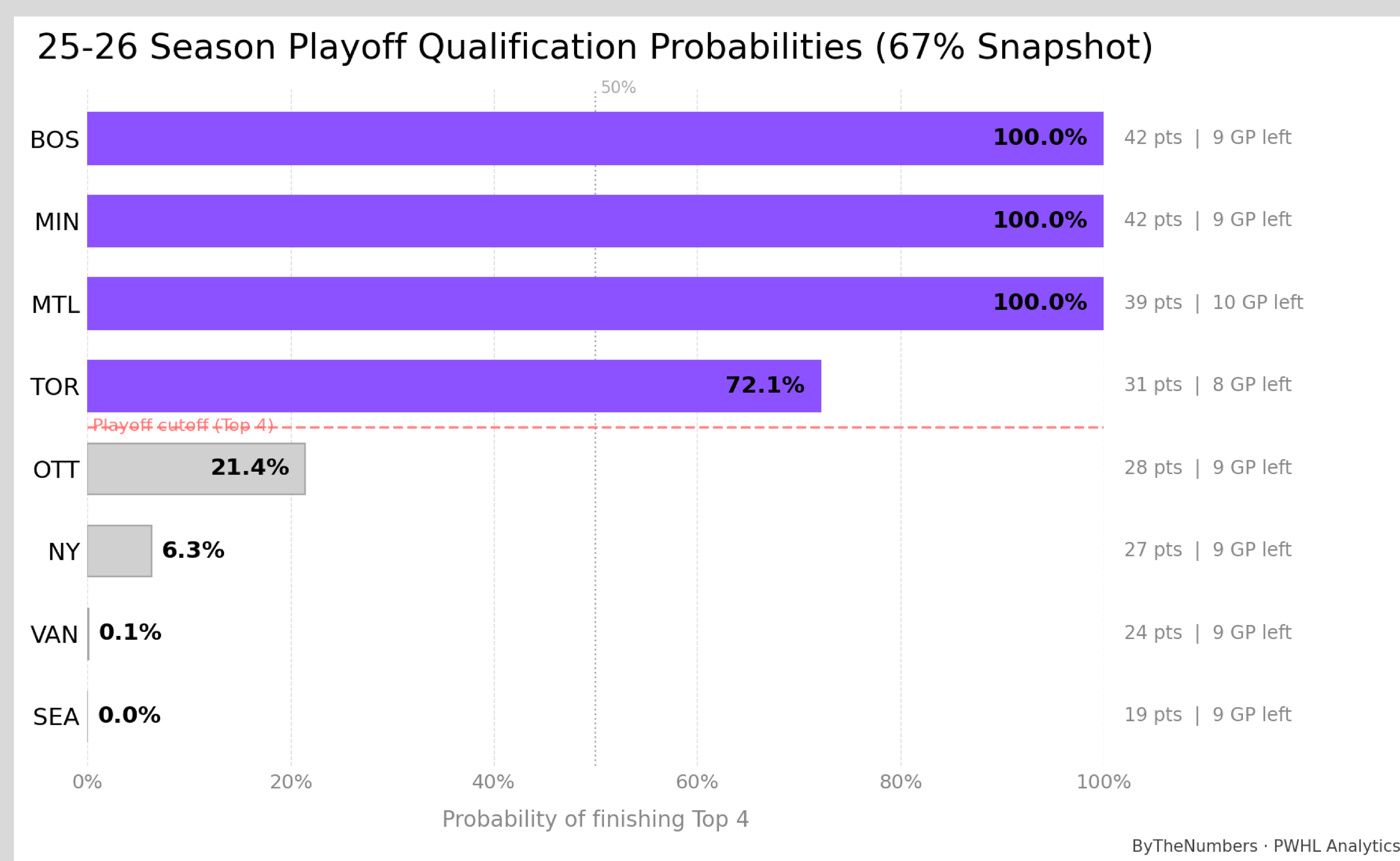
Playoff probabilities are estimated using **a Monte Carlo simulation** of the remaining schedule. Team scoring is modeled with **a Poisson process**, where expected goals are scaled from the league-average scoring rate based on relative team strength:

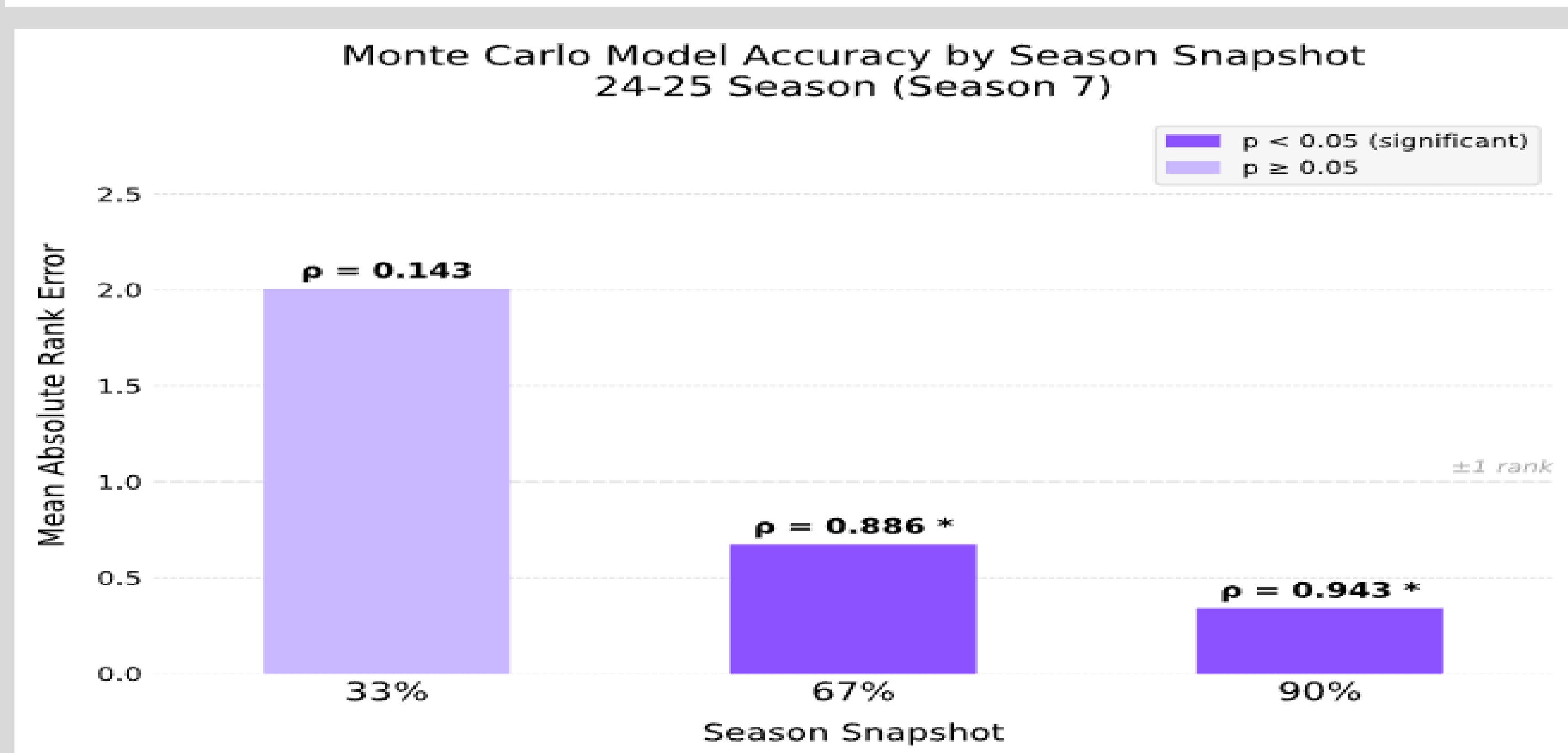
$$\lambda_{home} = 2.8 \cdot \frac{s_{home}}{\bar{s}} \cdot 1.08 \qquad \lambda_{away} = 2.8 \cdot \frac{s_{away}}{\bar{s}}$$

The model runs **10,000 simulated seasons**, with playoff probability defined as the proportion of simulations in which a team finishes in the top four of the standings. It is worth noting that with only six to eight teams in the league, **standing separation is sensitive to small point differentials**, which constrains the model's ability to discriminate mid-table teams this is a structural limitation of parity-driven emerging leagues rather than a modeling failure.

## Predictive Accuracy Improves After 67% of the Season

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## Conclusion/ Analysis

Model performance improved monotonically as more of the season was observed. In the 6-team 2024–2025 validation set, early-season predictions were unreliable, with Spearman correlation near zero at the **33% snapshot (ρ = 0.143, p = 0.787)** and remaining below statistical significance at **50% (ρ = 0.771, p = 0.072).** Predictive signal emerged at **the 67% snapshot (ρ = 0.886, p = 0.019)** and remained strong through the remainder of the season **(ρ = 0.943, p = 0.005).**

In the 8-team 2025–2026 validation set, the model achieved statistical significance earlier, **reaching ρ = 0.976 (p < 0.001) at the 67% snapshot** with perfect rank ordering. Across both seasons, these results **preliminarily suggest** that meaningful predictive signal emerges after approximately 67% of games have been played.

Comparison against an Elo baseline at the 67% snapshot shows comparable performance in 2024–2025 **(ρ = 0.943 vs. 0.943),** with the proposed model outperforming Elo in 2025–2026 **(ρ = 0.976 vs. 0.881).** The performance gap widening in the second season despite Elo having an additional season of rating history to draw from suggests that **explicit feature-based team strength modeling provides compounding cross-season advantages** over implicit rating accumulation as the league matures, though replication across additional seasons is needed to confirm this as a stable threshold.

Overall, these findings **support Monte Carlo simulation as a practical forecasting** approach for low-data sports environments, with reliability increasing as the season progresses with applications in real-time fan engagement, broadcast insights, and content generation.

## References


- **PWHL HockeyTech API** – Game statistics
- **LeagueStats** – Player images
- **Hvattum & Arntzen (2010)** – Using Elo ratings for match
- **Dixon & Coles 1997** – Ridge Regression and Poisson in Sports


Visualizations are shared via **BTN Sports**, reaching ~1–2K engagements per post and supporting fan-facing storytelling.